

If Dinosaurs Lived Today

A Climate-Based Species Distribution Model for Predicting Modern Habitat Suitability

Problem Statement

Which modern regions of Earth would be climatically suitable for a given dinosaur species, based on its estimated paleoclimate envelope and today's climate conditions? This project applies Species Distribution Modeling (SDM), an established ecological technique, to a paleontological question, combining paleoclimatology, GIS visualization, and machine learning.

Methodology

For each of 12 well-documented dinosaur species, a climate envelope (temperature, precipitation, humidity range) was reconstructed from published paleoclimate literature (leaf-margin analysis, CLAMP, paleosol geochemistry) where available. Each modern region's climate is scored against this envelope using a weighted Gaussian-falloff formula, then independently cross-validated against a real MaxEnt implementation (elapid), the algorithm class used in professional ecological SDM work.

$$\text{suitability} = 0.40 \times \text{temp_score} + 0.35 \times \text{precip_score} + 0.25 \times \text{humidity_score}$$

Key Findings

- 9 of 12 species now use citation-backed paleoclimate ranges from real published studies (Hell Creek, Morrison, Djadochta, and Bahariya Formation research), the remaining 3 use formation-level qualitative interpretation, flagged explicitly as a limitation.
- For Tyrannosaurus rex, both the Gaussian model and an independent real MaxEnt fit agree on the same top two modern regions (central Japan, Pacific Northwest USA), a meaningful cross-validation result.
- For Velociraptor, the model independently ranks the modern Gobi Desert near the top of its suitability list, consistent with published claims that the Djadochta Formation's paleoclimate closely resembles today's Gobi.

System & Tools

Python (pandas, NumPy, scikit-learn-adjacent elapid for MaxEnt), Plotly for geospatial visualization, Leaflet.js for the interactive web dashboard. Data sources: peer-reviewed paleoclimate literature (cited), NASA POWER API (free, no key) and the Paleobiology Database API (free, no key) for live data extension, both wired into ready-to-run Google Colab cells.

Limitations (stated explicitly, not hidden)

- No dinosaur climate model can be verified against ground truth; all outputs are estimates, not facts.
- MaxEnt validation currently uses synthetic presence points sampled from the estimated envelope, not real multi-locality fossil GPS coordinates (a ready-to-run PBDB live-fetch cell closes this gap).
- Modern climate data is a curated 59-region sample; a full global grid would need bulk gridded data (e.g. WorldClim).
- No food-web, competition, or human land-use modeling; diet-based survival modifier is illustrative only.

Deliverables

(1) Interactive multi-page web app (if_dinosaurs_lived_today.html) with a specimen catalog, per-species map dossier, and methodology page. (2) Executed research notebook with citations, live NASA POWER / PBDB fetch cells, and MaxEnt cross-validation. (3) Reusable Python scoring module and datasets.

Prepared as a portfolio / research-proposal supporting project. Demonstration-grade prototype; not a peer-reviewed paleoclimate reconstruction.